



Danburite

△ **Transparent**, single crystal of danburite

Danburite crystals are glassy prisms that resemble topaz, but in roughs they are distinguishable by their poor cleavage. Danburite is named after the city of Danbury, Connecticut, in the USA, where it was first discovered as a distinct species in 1839. It is usually colourless, but it can also be amber, yellow, grey, pink, or yellow-brown. Danburite is cut as a gemstone, both faceted and *en cabochon*, but is generally considered a collector's stone. Large gem danburites have been found in Dalnegorsk, Russia, in crystals up to 30cm (12in) long.

Specification

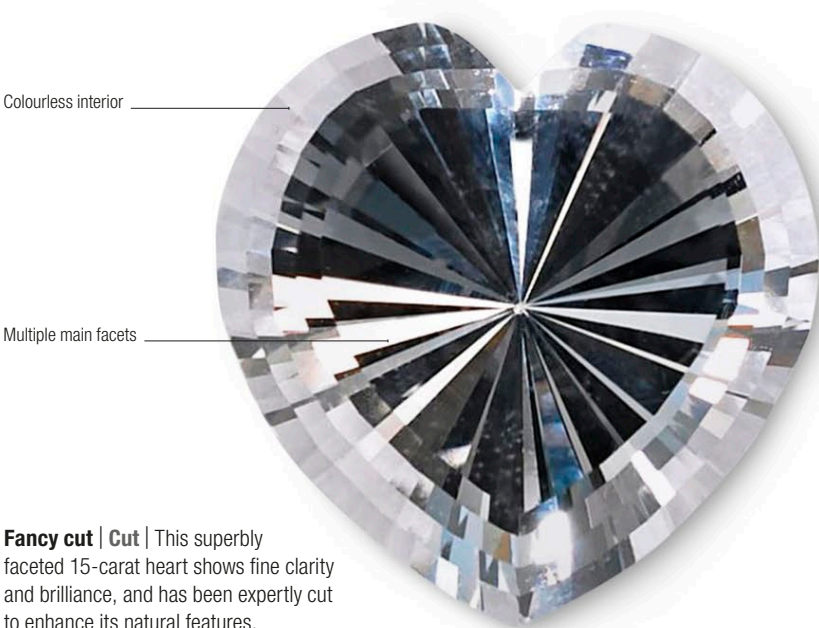
Chemical name	Calcium borosilicate	Formula	$\text{CaB}_2\text{Si}_2\text{O}_8$
Colours	Colourless, yellow, pink, yellow-brown	Structure	Orthorhombic
Hardness	7–7.5	SG	3
RI	1.63–1.64	Lustre	Vitreous to greasy
Streak	White	Locations	Switzerland, Russia, Myanmar, Slovakia, USA, Mexico, Madagascar



Yellow rough | Rough | This stream-rounded rough weighs around 23 carats. It comes from Tanzania, where a large amount of yellow material has been found.



Mexican danburite | Rough | These white crystals from San Luis Potosi, Mexico, are in parallel growth and show perfect prismatic form and classic terminations.



Fancy cut | Cut | This superbly faceted 15-carat heart shows fine clarity and brilliance, and has been expertly cut to enhance its natural features.

The Danbury mineral zone

A New England gem belt

The small town of Danbury, in Fairfax County, Connecticut, USA, is the centre of a local mineral belt running for several kilometres, which has produced over 50 different mineral types. It is an area of complex geological structure, with folding, faulting, and metamorphism, resulting in its rich mineralogy. As well as producing danburite, the area has also been an important source of zircon, baryte, celestine, moonstone, sphene, diopside, rutile, garnet, quartz, and pyrite.



Danburite crystal The Connecticut town of Danbury lends its name to the mineral that was discovered there.



Round brilliant | Cut | Danburite often rivals topaz in its brilliance and clarity, as can be seen in this flawless, round brilliant-cut gemstone.



Mixed earrings | Set | This earring set's pair of faceted danburites from Mexico contrasts with its pink tourmaline cabochons from Myanmar.